ROBERT v 1.0.5 2026/04/20 12:21:15

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## ROBERT SCORE

*This score is designed to analyze the predictive ability of the models using different metrics.*

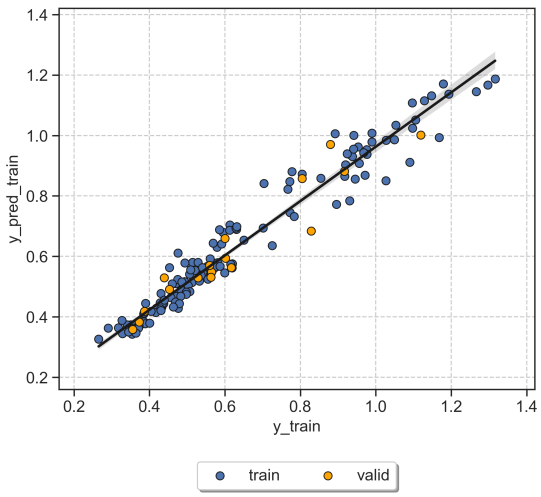
**No PFI (all descriptors):**

ML model: RF
Proportion Train:Validation = 90:10

STRONG

**The model has a score of 9/10**

- ●● The valid. set shows an $R^2$ of 0.91
- ●● The valid. set has 5.9% of outliers
- ●● Using 166:8 points(train+valid.):descriptors
- ●●● The valid. set passes 3 VERIFY tests




Train : $R^2$ = 0.96, MAE = 0.038, RMSE = 0.055
Valid. : $R^2$ = 0.91, MAE = 0.047, RMSE = 0.062

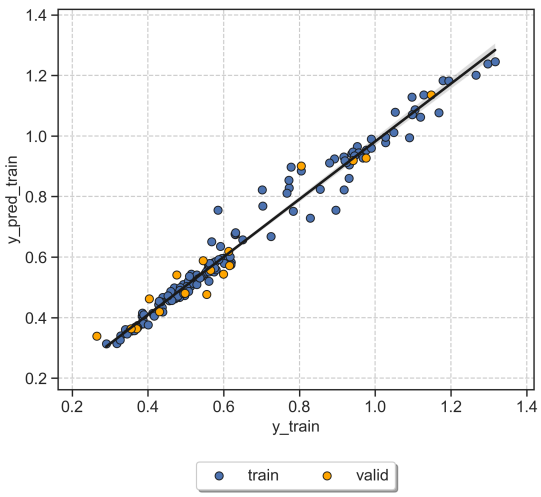
**PFI (only important descriptors):**

ML model: RF
Proportion Train:Validation = 90:10

STRONG

**The model has a score of 9/10**

- ●● The valid. set shows an $R^2$ of 0.96
- ●● The valid. set has 5.9% of outliers
- ●● Using 166:4 points(train+valid.):descriptors
- ●●● The valid. set passes 3 VERIFY tests




Train : $R^2$ = 0.98, MAE = 0.024, RMSE = 0.038
Valid. : $R^2$ = 0.96, MAE = 0.039, RMSE = 0.048

**Score thresholds** *(detailed in https://robert.readthedocs.io/en/latest/Score/score.html)*

| $R^2$ | Outliers | Points:descriptors | VERIFY tests |
|---|---|---|---|
| ●● $R^2$ > 0.85 | ●● < 7.5% of outliers | ●● > 10:1 p:d ratio | Up to ●●●● (tests pass) |
| ● 0.85 > $R^2$ > 0.70 | ● 7.5% < outliers < 15% | ● 10:1 > p:d ratio > 3:1 | - (all tests failed) |
| - $R^2$ < 0.70 | - > 15% of outliers | - p:d ratio < 3:1 | |

Some tips to improve the score

✓ A ROBERT score of 9 or 10 suggests that the predictive ability of your model is strong, congratulations!

How to predict new values with these models?

1. Create a CSV database with the new points, including the necessary descriptors.

2. Place the CSV file in the parent folder (i.e., where the module folders were created)

3. Run the PREDICT module as 'python -m robert --predict --csv_test FILENAME.csv'.

4. The predictions will be shown at the end of the resulting PDF report and will be stored in the last column of two CSV files called MODEL_SIZE_test(_No)_PFI.csv, which are in the PREDICT folder.

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## REPRODUCIBILITY

*This section provides all the instructions to reproduce the results presented.*

**1. Download these files *(the authors should have uploaded the files as supporting information!)*:**

- CSV database (data_180_days_all_cells.csv)

**2. Install and adjust the versions of the following Python modules:**

- Install ROBERT and its dependencies: conda install -c conda-forge robert
- Adjust ROBERT version: pip install robert==1.0.5
- scikit-learn-intelex: not installed

*(if scikit-learn-intelex is installed, slightly different results might be obtained)*

**3. Run ROBERT using this command line in the folder with the CSV database:**

python -m robert --model "[RF]" --train "[90]" --names "index" --ignore "[index]" --y "PCE" --auto_test "False" --csv_name "data_180_days_all_cells.csv"

**4. Execution time, Python version and OS:**

Originally run in Python 3.9.25 using Windows 10.0.19045

Total execution time: 21.19 seconds *(the number of processors should be specified by the user)*

## TRANSPARENCY

*This section contains important parameters used in scikit-learn models and ROBERT.*

**1. Parameters of the scikit-learn models (same keywords as used in scikit-learn):**

**No PFI (all descriptors):**

sklearn model: RandomForestRegressor
random_state: 8
names: index
n_estimators: 20
max_depth: 100
max_features: 1.0
min_samples_split: 2
min_samples_leaf: 1
min_weight_fraction_leaf: 0
ccp_alpha: 0
oob_score: False
max_samples: 0.5

**PFI (only important descriptors):**

sklearn model: RandomForestRegressor
random_state: 70
names: index
n_estimators: 40
max_depth: 20
max_features: 1.0
min_samples_split: 2
min_samples_leaf: 1
min_weight_fraction_leaf: 0
ccp_alpha: 0
oob_score: False
max_samples: 0.75

**2. ROBERT options for data split (KN or RND), predict type (REG or CLAS) and hyperopt error (RMSE, etc.):**

**No PFI (all descriptors):**

split: RND
type: reg
error_type: rmse

**PFI (only important descriptors):**

split: RND
type: reg
error_type: rmse

## ABBREVIATIONS

*Reference section for the abbreviations used.*

**ACC:** accuracy
**ADAB:** AdaBoost
**CSV:** comma separated values
**CLAS:** classification
**CV:** cross-validation
**F1 score:** balanced F-score
**GB:** gradient boosting
**GP:** gaussian process
**KN:** k-nearest neighbors
**MAE:** root-mean-square error
**MCC:** Matthew's correl. coefficient
**ML:** machine learning
**MVL:** multivariate lineal models
**NN:** neural network
**PFI:** permutation feature importance
**R2:** coefficient of determination
**REG:** Regression
**RF:** random forest
**RMSE:** root mean square error
**RND:** random
**SHAP:** Shapley additive explanations
**VR:** voting regressor

## CURATE

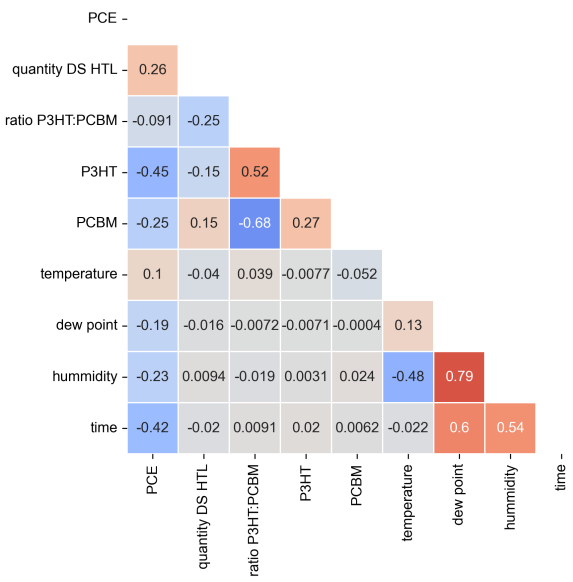
*This module takes care of data curation, including filters for correlated descriptors, noise, and duplicates, as well as conversion of categorical descriptors.*

The complete output (CURATE_data.dat) and curated database are stored in the CURATE folder.

Time CURATE: 0.47 seconds

**------- Images generated by the CURATE module -------**

**Pearson's r heatmap**



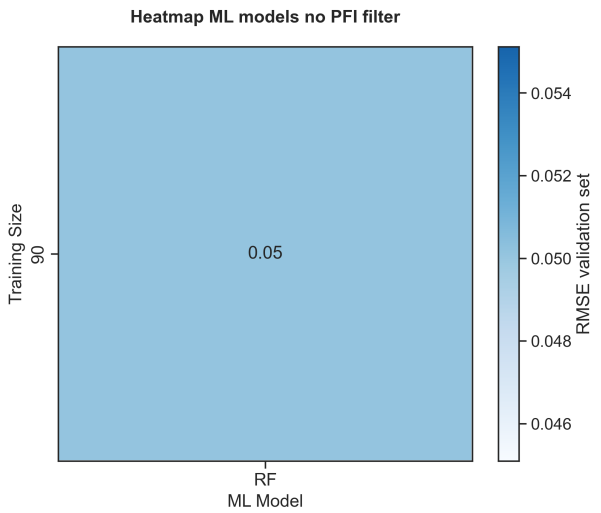

## GENERATE

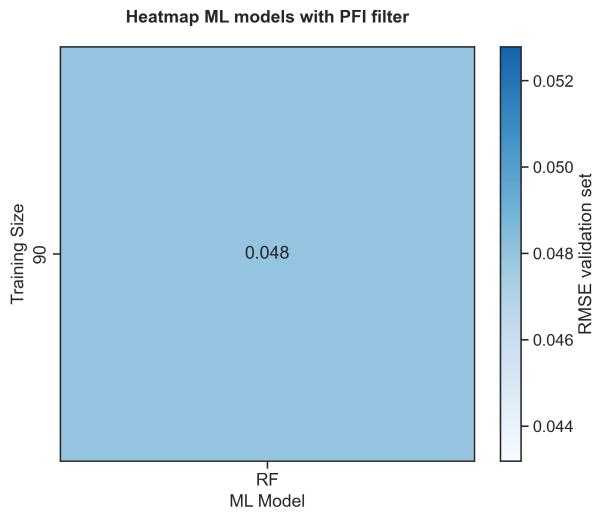
*This module carries out a screening of ML models and selects the most accurate one. It includes a comparison of multiple hyperoptimized models and training sizes.*

The complete output (GENERATE_data.dat) and heatmaps are stored in the GENERATE folder.

Time GENERATE: 13.69 seconds

**------- Images generated by the GENERATE module -------**




## VERIFY

*Determination of predictive ability of models using four tests: 5-fold CV, y-mean (error against the mean y baseline), y-shuffle (predict with shuffled y values), and one-hot (predict using one-hot encoding instead of the X values).*

The complete output (VERIFY_data.dat) and donut plot are stored in the VERIFY folder.

Time VERIFY: 1.38 seconds

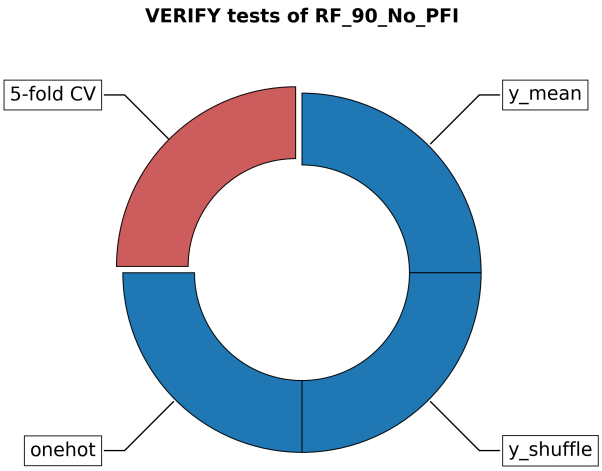
**------- Images and summary generated by the VERIFY module -------**

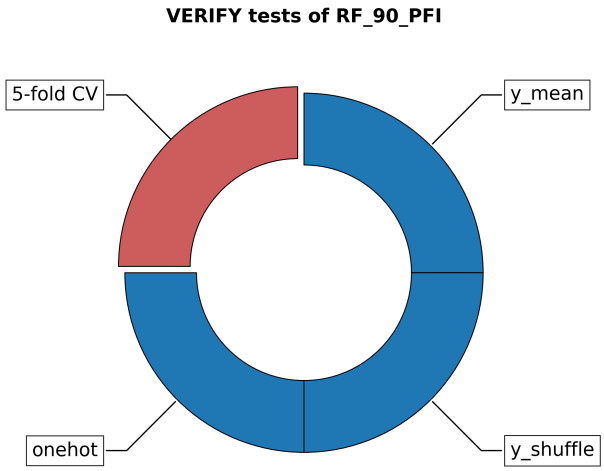
**No PFI (all descriptors):**

Original RMSE (valid. set) 0.062 + 25% thres. = 0.078
- x 5-fold CV: FAILED, RMSE = 0.096, higher than thres.
- o y_mean: PASSED, RMSE = 0.21, higher than thres.
- o y_shuffle: PASSED, RMSE = 0.3, higher than thres.
- o onehot: PASSED, RMSE = 0.2, higher than thres.

**PFI (only important descriptors):**

Original RMSE (valid. set) 0.048 + 25% thres. = 0.06
- x 5-fold CV: FAILED, RMSE = 0.078, higher than thres.
- o y_mean: PASSED, RMSE = 0.23, higher than thres.
- o y_shuffle: PASSED, RMSE = 0.32, higher than thres.
- o onehot: PASSED, RMSE = 0.24, higher than thres.



## PREDICT

*This module predicts and plots the results of training and validation sets from GENERATE, as well as from external test sets (if any). Feature importances from SHAP and PFI, and outlier analysis are also represented.*

The complete output (PREDICT_data.dat) and heatmaps are stored in the PREDICT folder.

Time PREDICT: 5.65 seconds

**------- Images and summary generated by the PREDICT module -------**

**No PFI (all descriptors):**

Prediction metrics and descriptors
- Points Train:Validation = 149:17
- Proportion Train:Validation = 90:10
- Number of descriptors = 8
- Proportion (train+valid.) points:descriptors = 166:8
- Train : $R^2$ = 0.96, MAE = 0.038, RMSE = 0.055
- Valid. : $R^2$ = 0.91, MAE = 0.047, RMSE = 0.062

Outliers (max. 10 shown)
Train: 10 outliers out of 149 datapoints (6.7%)
- 1 (3.4 SDs)
- 3 (2.3 SDs)
- 4 (2.1 SDs)
- 5 (3.5 SDs)
- 8 (3.5 SDs)
- 9 (2.1 SDs)
- 13 (2.7 SDs)
- 67 (2.4 SDs)
- 72 (2.3 SDs)
- 163 (2.5 SDs)

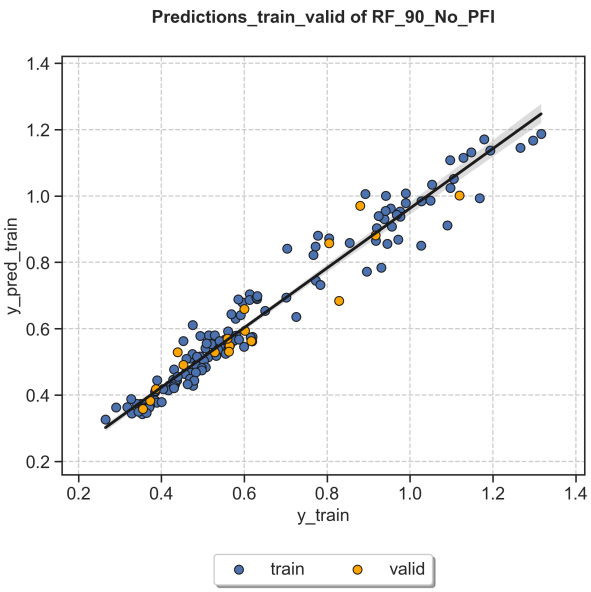
Validation: 1 outliers out of 17 datapoints (5.9%)
- 14 (2.7 SDs)

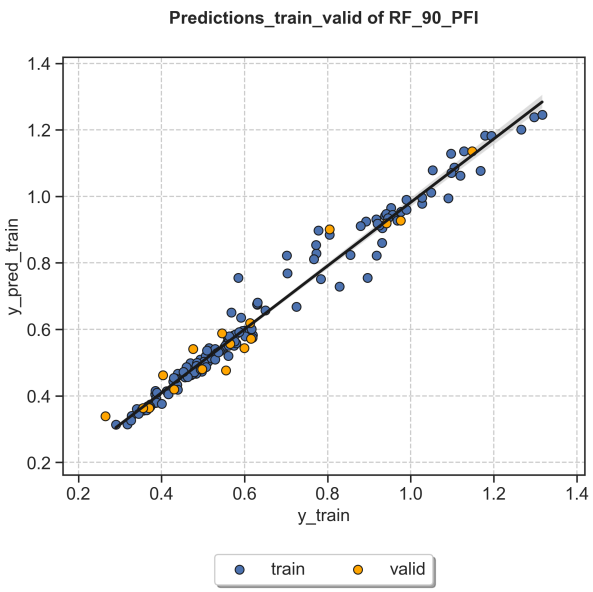
**PFI (only important descriptors):**

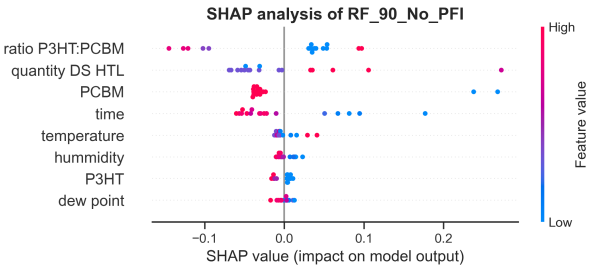
Prediction metrics and descriptors
- Points Train:Validation = 149:17
- Proportion Train:Validation = 90:10
- Number of descriptors = 4
- Proportion (train+valid.) points:descriptors = 166:4
- Train : $R^2$ = 0.98, MAE = 0.024, RMSE = 0.038
- Valid. : $R^2$ = 0.96, MAE = 0.039, RMSE = 0.048

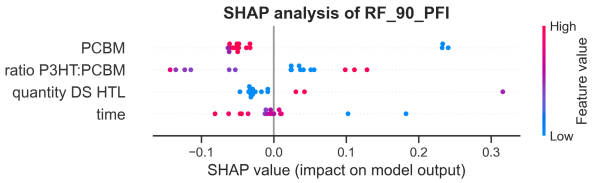
Outliers (max. 10 shown)
Train: 8 outliers out of 149 datapoints (5.4%)
- 1 (2.3 SDs)
- 2 (2.4 SDs)
- 5 (2.4 SDs)
- 9 (4.0 SDs)
- 14 (2.6 SDs)
- 16 (3.3 SDs)
- 70 (3.3 SDs)
- 89 (5.0 SDs)

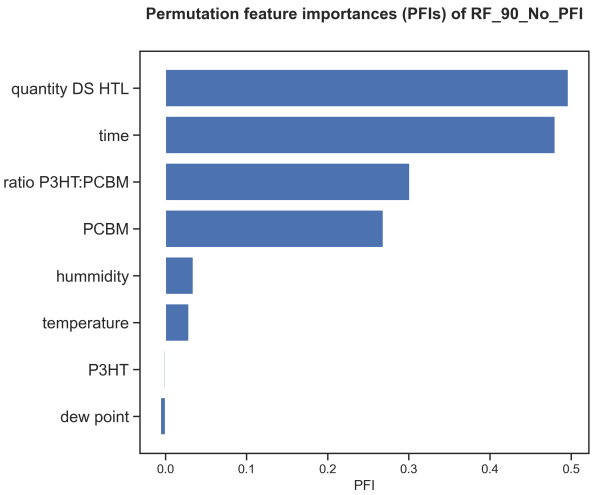
Validation: 1 outliers out of 17 datapoints (5.9%)
- 149 (2.5 SDs)

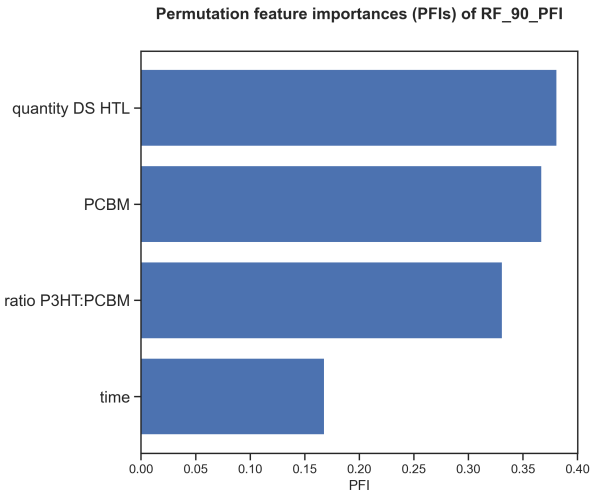

Predictions_train_valid of RF_90_No_PFI
y_pred_train
y_train
train
valid

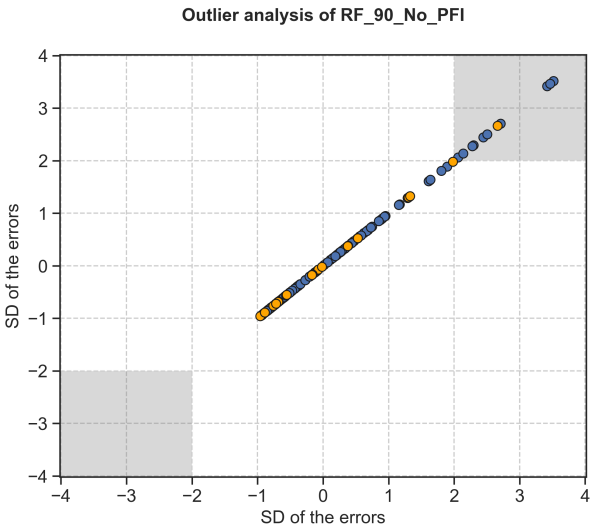

Predictions_train_valid of RF_90_PFI
y_pred_train
y_train
train
valid

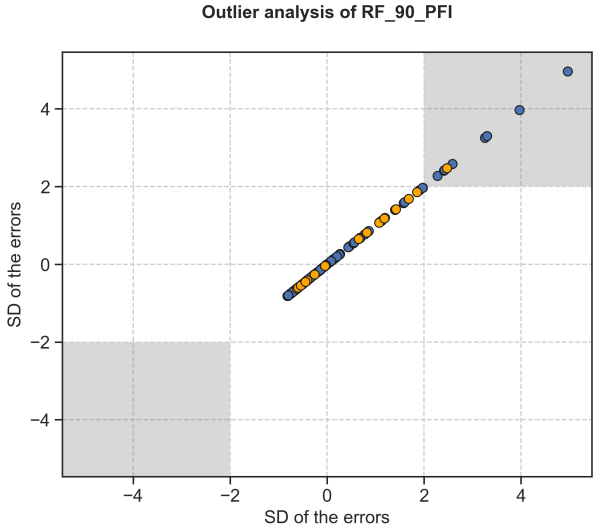

SHAP analysis of RF_90_No_PFI
ratio P3HT:PCBM
quantity DS HTL
PCBM
time
temperature
hummidity
P3HT
dew point
SHAP value (impact on model output)
High
Feature value
Low


SHAP analysis of RF_90_PFI
PCBM
ratio P3HT:PCBM
quantity DS HTL
time
SHAP value (impact on model output)
High
Feature value
Low


Permutation feature importances (PFIs) of RF_90_No_PFI
quantity DS HTL
time
ratio P3HT:PCBM
PCBM
hummidity
temperature
P3HT
dew point
PFI


Permutation feature importances (PFIs) of RF_90_PFI
quantity DS HTL
PCBM
ratio P3HT:PCBM
time
PFI


Outlier analysis of RF_90_No_PFI
SD of the errors
SD of the errors
4
3
2
1
0
−1
−2
−3
−4
−4
−3
−2
−1
0
1
2
3
4


Outlier analysis of RF_90_PFI
SD of the errors
SD of the errors
4
2
0
−2
−4
−4
−2
0
2
4